weapons. Presently, various technological tools and techniques have been deployed to combat terrorism. Some of these technologies are fairly mature, while others show great potential but still require some years of research and development before these are fully operational.

Yet, all technologies share a common characteristic; these offer significant potential solutions to address the most pressing antiterrorism concerns.

The technologies are:

- Directed-energy weapons
- Non-lethal weapons
- Nanotechnology
- Biometrics
- Data mining and analysis technologies
- Network-centric operations

Directed-energy weapons. These weapons generate very high-power beams such as lasers and microwave radiations that are precisely focused to hit targets with sub-atomic particles, both to track and destroy.

Directed-energy weapons have the capability to cause casualties, damage equipment, disable targets on ground, air and sea, and provide active defence against threats from artillery, rockets, mortars, missiles and unmanned aerial vehicles.

Non-lethal weapons. These are employed to incapacitate personnel or material while minimising fatalities and undesired damage to property and environment. Their functions include preventing or neutralising the means of transportation such as vehicles, vessels or aircraft including those for weapons of mass destruction.

The technologies used include acoustics systems, chemicals, communications systems, electromagnetic and electrical systems, entanglement and other mechanical systems, information technologies, optical devices, non-penetrating projectiles and munitions, and more. These can be integrated with systems to make these more effective and discriminate.

Well-known terrorist attacks in recent years

September 11, 2001, the USA. Four coordinated terrorist attacks by Islamic terrorist group *Al-Qaeda* on the USA killed 2996 people, injured more than 6000 others and caused at least US$ 10 billion in property and infrastructure damage and US$ 3 trillion in total costs.

July 13, 2011, Mumbai, India. A series of three coordinated bomb explosions took place at different locations, namely, Opera House, Zaveri Bazaar and Dadar West localities, between 18:54 and 19:06 IST, leaving 26 killed and 130 injured.

November 13-14, 2015, Paris. Multiple shootings and grenade attacks at a music venue, sports stadium and several bar and restaurant terraces lead to the killing of 130 people, and injuring 352. The ISIS claimed responsibility for the attacks.

Fig. 2: A laser with US Naval Sea Systems Command (NSSC), which is an electromagnetic gun prototype (Image courtesy: www.occupycorporatism.com)

Nanotechnology. Nanotechnology involves developing materials and complete systems at atomic, molecular or macromolecular levels where dimensions fall in the range of one to 100 nanometres. Fabrication at such nano scale offers unique capabilities, and materials can be made to have specific properties.

Among antiterrorism applications, sensors using nanotechnology are most important. Nano-scale sensors form a weak chemical bond with the substance and then change their properties in response (such as colour change, or a change in conductivity, fluorescence or weight). As an antiterrorism tool, nanotechnologies are relatively in their infancy stage *vis-à-vis* other technologies.

Biometrics. This refers to recorded unique physical or behavioural characteristics of individuals. These are more reliable and more difficult to forget, lose, get stolen, falsified or be guessed. Biometrics can be used for verification or identification. For identification, a person's presented biometric is compared with all biometric templates within a database.

Five major types of biometric technologies available today are:

1. Iris recognition that relies on distinctly-coloured ring surrounding the pupil of the eye

2. Hand geometry that relies on measurements of fingers, distances between joints and the shapes of knuckles

3. Fingerprint recognition that re-

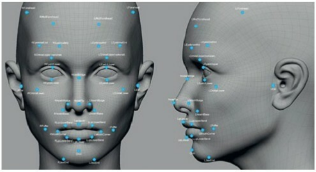

Fig. 3: Face recognition technology identifies individuals by analysing several features such as the upper outlines of eye sockets and sides of the mouth (Image courtesy: www.extremetech.com)

Fig. 4: Al-Qaida created by Osama Bin Laden relies heavily on IT (Image courtesy: www.independent.co.uk)